



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

CONCRETE WORKS

BDCCW401

Perform Concrete works

Competence

RQF Level: 4

Learning Hours



120

Credits: 12

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 In Building Construction

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Purpose statement	This module describes the skills, knowledge and attitude required to perform concrete works on construction site. It is intended to be pursued by learners at TVET level IV in Building Construction. At the end of this module, the learner will be able to: erect formwork, mix concrete, cast concrete, dismantle the formwork as required and clean the workplace.					
Leaning assumed to be in place	Building construction drawing, Basic welding					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Perform preliminary works.	1.1. PPE is properly used according to the work to be done.
	1.2. Working /casting area is properly prepared according to the work to be done.
	1.3. Tools are appropriately selected in accordance to the work to be done.
	1.4. Equipment is appropriately selected and used in accordance to the work to be done.
	1.5. Materials are appropriately selected according to the work to be done.
	1.6. Concrete spacers are properly prepared according to the designed concrete cover.
2. Mix concrete.	2.1. Mixing ratio is properly identified according the designed concrete class/grade.
	2.2. Concrete materials are properly quantified based on given mixing ratio.
	2.3. Mixing place is adequately prepared as per required standards.
	2.4. Concrete materials are properly collected according to their types.

	2.5. Concrete materials are accurately batched in accordance with mix ratio.
	2.6. Concrete materials are correctly mixed according to the mixing methods.
3. Erect formwork.	3.1. Type of formwork is correctly identified according to the material.
	3.2. Formwork pieces are correctly prepared according to the work to be done.
	3.3. Formwork panels are properly prepared according to the structure element to be made.
	3.4. Formwork panels are properly assembled according to the structure element to be made.
	3.5. Formwork supporting structure is accurately prepared according to the structure element to be made.
	3.6. Formwork is tightly fixed according to the designed structure.
4. Cast concrete.	4.1. Formwork is properly assessed before concrete casting.
	4.2. Concrete is properly transported in accordance with transport modes.
	4.3. Concrete is properly poured and vibrated according to the standards.
	4.4. Concrete surface is properly leveled according to the design.
	4.5. Concrete tests are properly carried out according to the standards.
	4.6. Concrete is adequately cured according to the standards.
5. Clean the workplace.	5.1. Working area is properly cleaned after work.
	5.2. Used tools and equipment are properly cleaned according to their types.
	5.3. Used tools and equipment are correctly stored in their right place.
6. Dismantle the formwork.	6.1. Dismantling tools are properly prepared according to their use.
	6.2. Formwork panels are safely disassembled according to the standards.

	6.3. Disassembled formwork materials are adequately arranged according to their type.
	6.4. Workplace is properly cleaned after work as per type of dirt's cleaning methods.
	6.5. Used tools and equipment are properly collected and stored according to their category and status.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify Concrete materials, Tools and Equipment ✓ Identify Cleaning methods for concrete mixing and casting area ✓ Select Concrete ingredients Batching and mixing methods ✓ Identify Characteristics of good concrete ✓ Identify concrete tests ✓ Identify Concrete defects and different solutions ✓ Identify different building elements ✓ Identify tools and equipment used for formwork making	✓ Apply process of concrete ingredients mixing ✓ Test concrete ✓ Apply concrete casting and vibrating technics ✓ Apply Curing process ✓ Apply Tools, Equipment and material handling technics ✓ Apply different Formwork assembling techniques ✓ Take measurements ✓ Use hand tools, power tools and equipment to cut and shape formwork materials ✓ Assemble and support formwork panels ✓ Estimate quantity and cost of concrete works ✓ Manage work and personal at work	✓ Demonstrate an understanding of workplace hazards and comply with safety measures ✓ Communicate and negotiate effectively at work with clients and team members ✓ Demonstrate pressure tolerance ✓ Demonstrate optimism and reliability during work execution ✓ Pay attention to details ✓ Demonstrate persistent, team working, flexibility, and punctuality at work

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform preliminary works. 2. Mix concrete. 3. Erect formwork. 4. Cast concrete. 5. Clean the workplace. 6. Dismantle the formwork.
Learning outcome 1: Perform preliminary works.	Learning hours: 15
Indicative content	
• Personal Protective Equipment for concrete works ➤ Types of PPE for concrete works. ✓ Overall ✓ Strong boots ✓ Helmet ✓ Gloves ✓ Goggles ✓ Face mask ➤ Functions of PPE ➤ Advantages & Disadvantages • Working area preparation ➤ Techniques of working area cleaning. ➤ Benefits of working area cleaning. • Tools used in concrete works. ➤ Types of tools used in concrete works. ➤ Use/function of tools for concrete works. ➤ Maintenance of tools. • Equipment used in concrete works. ➤ Types of equipment used in concrete works. ➤ Function of equipment for concrete works. ➤ Maintenance of equipment • Materials used in concrete works. ➤ Types of materials.	

- Functions of materials.
- Qualities of concrete materials.
- Properties of concrete materials.

- **Concrete spacers preparation.**

- Description of concrete spacers.
 - ✓ Definition of concrete spacers
 - ✓ Function of concrete spacers.
 - ✓ Standards sizes of concrete spacers.
 - ✓ Classification of concrete spacers.
- Production of precast concrete spacers.

Resources required for the learning outcome

Equipment	PPEs, wheelbarrow, tipper trucks, air compressor, concrete pump, beam balance, gauge box, concrete mixer, poker vibrator, power float, dumpy level, belt conveyor, chute, truck mixer, crane, batching plant, metallic/plastic pans, metallic/plastic buckets, jerrycan, demolition hammer, electric circular saw, electric plane machine, miter saw machine, welding machine.
Materials	Building lines, timber, water, Fuel, Fine and coarse aggregates, cement, nails, pegs, warning tape, admixtures, binding wire, ply wood, steel, mild steel sheets aluminum sheets, magnesium boards, screws, drill bits, welding rod (electrode), lubricants, chalks, plastic sheeting.
Tools	Wire brushes, straight edge, spade/shovel, hoe, pick axe, machete / panga, trowels, spirit level, claw hammer, club hammer, squares, marker pens, chisel, hack saw, tape measure, portable circular saw, jack plane, plumb bob, clamps, screwing machines, drilling machines, claw bar, nail puller, backhoe, broom, float, hose pipe.
Facilitation techniques	• Brainstorming • Group discussion • Individualized • Trainer guided • Demonstration and simulation • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion

- Open ended questions
- Performance assessment
- Product assessment

Learning outcome 2: Mix concrete.

Learning hours: 15

Indicative content

- **Concrete mixing ratio.**
 - Definition of mixing ratio.
 - Standards concrete mixing ratios.
 - Concrete classes/ Grades.
 - Areas of application of different concrete mixing ratios.
- **Concrete materials quantification.**
 - Methods of concrete material quantification.
 - ✓ Arbitrary method.
 - ✓ Wet & dry volume mix method.
- **Mixing place preparation**
 - Requirements of mixing place.
 - Mixing place preparation steps.
- **Concrete materials collection.**
 - Tips of concrete materials arrangement.
 - Storage of concrete material.
 - ✓ Internal storage
 - ✓ External storage
- **Batching of concrete materials**
 - Meaning of batching.
 - Importance of batching in concrete.
 - ✓ Consistency and uniformity
 - ✓ Cost-effectiveness
 - ✓ Quality control
 - Methods of batching.
 - ✓ Volume batching
 - ✓ Weight batching
 - Concrete admixtures:

✓ Types of concrete admixtures ✓ Functions of concrete admixtures • Mixing of concrete. ➤ Methods of concrete mixing. ✓ Hand/manual mixing method. ✓ Machine/Mechanical mixing method. ➤ Application of mixing methods ✓ Hand mixing procedure. ✓ Machine mixing procedure.
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Resources required for the indicative content

Equipment	PPEs, wheelbarrow, tipper trucks, air compressor, beam balance, gauge box, concrete mixer, batching plant, metallic/plastic pans, metallic/plastic buckets, jerrycan.
Materials	Water, Fuel, Fine and coarse aggregates, cement, warning tape, admixtures , bricks, Stones.
Tools	Spade/shovel, hoe, pick axe, machete / panga, trowelsclub hammer, squares, saw, tape measure, broom, float.
Facilitation techniques	• Group discussion • Brainstorming • Individualized • Trainer guided • Demonstration and simulation • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Open ended questions • Performance assessment • Product assessment

Learning outcome 3: Erect formwork.	Learning hours: 50
Indicative content	

- **Formwork identification.**

- Definition.
- Types of formworks based on material.
 - ✓ Timber formwork.
 - ✓ Plywood formwork.
 - ✓ Metallic formwork.
 - ✓ Plastic formwork.
 - ✓ Stay-in place formwork.
 - ✓ Permanent insulated formwork.
- Types of formworks based on the structure element.
 - ✓ Footing formwork.
 - ✓ Column formwork.
 - ✓ Beam formwork.
 - ✓ Slab formwork.
 - ✓ Stair formwork.
 - ✓ Wall formwork.
 - ✓ Ramp formwork.
- Characteristics of good formwork.

- **Operations of preparing formwork pieces**

- Planning
- Measure and mark the piece.
- Cutting the piece to size.
- Operating.
- Finishing /application of lubricants.

- **Formwork panels preparation.**

- Techniques of assembling prepared pieces to size.
 - ✓ Nailing
 - ✓ Bolt and nut
 - ✓ Wire tightening
 - ✓ Welding
 - ✓ Screwing
- Apply internal surface treatment.
 - ✓ Lubrication
 - ✓ Application of sheeting.

- **Formwork panels assembling process.**

- Setting out of formwork structure element (Footing, Column, Beam, Slab, Stair, Wall, Ramp)
- Formwork panels assembling techniques.

- **Formwork supporting and fixing.**

- Creating supporting structure.
 - ✓ By strapping of steel bars.
 - ✓ By timber stands/ Supports.
- Aligning and levelling of formwork techniques.
 - ✓ By spirit level.
 - ✓ By hose pipe.
 - ✓ By plumb bob.
 - ✓ By cords.
 - ✓ By levelling instruments.
- Fixing the formwork.
- Formwork finishing.
 - ✓ Sealing the joints.
 - ✓ Formwork cleaning.

Resources required for the indicative content

Equipment	PPEs, wheelbarrow, dumpy level, electric circular saw, electric plane machine, miter saw machine and welding machine.
Materials	Building lines, timber, nails, pegs, warning tape, ply wood, steel, mild steel sheets aluminum sheets, magnesium boards, screws, drill bits, welding rod (electrode), lubricants, chalks and plastic sheeting.
Tools	Wire brushes, straight edge, pick axe, machete / panga, trowels, spirit level, claw hammer, squares, marker pens, chisel, hack saw, tape measure, portable circular saw, jack plane, plumb bob, clamps, screwing machines, drilling machines, claw bar, and hose pipe.
Facilitation techniques	• Brainstorming • Group discussion • Individualized • Trainer guided • Demonstration and simulation • Individual and group work • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Open ended questions

- Performance assessment
- Product assessment

Learning outcome 4: Cast concrete.

Learning hours: 25

Indicative content

- **Formwork assessment points:**

- Checking verticality.
- Checking horizontality.
- Checking alignment.
- Checking rigidity and stability.
- Checking for leakage.

- **Concrete transportation modes.**

- Manual transportation.
 - ✓ By pans and buckets.
 - ✓ By wheelbarrow.
- Machine transportation.
 - ✓ By chute.
 - ✓ By pumps.
 - ✓ By belt conveyor.
 - ✓ By truck mixer.
 - ✓ By crane.
- Effects of poor transportation.
 - ✓ Segregation.
 - ✓ Bleeding.
 - ✓ Laitance.

- **Concrete pouring.**

- Pouring methods.
- Concrete vibration techniques.
 - ✓ Internal
 - ✓ External
- Concrete surface levelling.

- **Concrete testing.**

- Introduction to concrete properties.

- ✓ For fresh concrete.
- ✓ For hardened concrete.
- Concrete defects.
- Concrete tests.
 - ✓ Non-destructive test (rebound hammer).
 - ✓ Ultrasonic pulse velocity test.
 - ✓ Pull out test.
 - ✓ Penetration resistance test.
 - ✓ Setting time test by Vicat apparatus.

- **Concrete curing.**

- Curing techniques
- Curing process.

Resources required for the indicative content

Equipment	PPEs, wheelbarrow, tipper trucks, air compressor, concrete pump, concrete mixer, poker vibrator, power float, dumpy level, belt conveyor, chute, truck mixer, crane, batching plant, metallic/plastic pans, metallic/plastic buckets, jerrycan.
Materials	water, Fuel, cement, warning tape, admixtures
Tools	Wire brushes, straight edge, spade/shovel, hoe, machete / panga, trowels, spirit level, claw hammer, broom, float and hose pipe.
Facilitation techniques	• Group discussion • Individualized • Trainer guided • Demonstration and simulation • Individual and group work • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Open ended questions • Performance assessment • Product assessment

Learning outcome 5: Clean the workplace.

Learning hours: 5

Indicative content

- **Cleaning of working area**
 - Identification of unwanted materials
 - Remove unwanted material
- **Cleaning of tools and equipment**
 - Cleaning methods
 - Apply methods of cleaning tools and equipment
- **Storing tools and equipment**
 - Storing procedure
 - Safe storing of tools and equipment

Resources required for the indicative content

Equipment	PPEs, wheelbarrow, jerrycan, bucket and air compressor.
Materials	Water, Fuel, lubricants.
Tools	Wire brushes, spade/shovel, hoe, trowels and broom.
Facilitation techniques	• Group discussion • Individualized • Trainer guided • Demonstration and simulation • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions • Performance assessment • Product assessment

Learning outcome 6: Dismantle the formwork.

Learning hours: 10

Indicative content

- **Formwork dismantling tools and equipment**
 - Sledge hammer
 - Claw bar
 - Nail puller
 - Demolition hammer

➤ Backhoe

- **Formwork dismantling process**

- Prepare dismantling tools and equipment
- Remove the stakes form supporting structure
- Disassemble panels

- **Arrangement of disassembled formwork parts**

- **Workplace cleaning**

- **Procedure of storing tools and equipment**

Resources required for the indicative content

Equipment	PPEs, wheelbarrow, air compressor, jerrycan and demolition hammer
Materials	Water, Fuel, Lubricants.
Tools	Wire brushes, claw hammer, club hammer, chisel, hack saw, claw bar, nail puller, backhoe and broom.
Facilitation techniques	• Group discussion • Individualized • Trainer guided • Demonstration and simulation • Individual and group work • Practical exercise
Formative assessment methods	• Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions • Written assessment • Performance assessment • Product assessment

Integrated/Summative assessment (For specific module)

Integrated situation

MUHABURA Construction Company Ltd is a company that is executing a construction project of commercial building in Nyarugenge District Muhima sector. The work progress is under starting stage where setting out and excavation for foundation have been completed but the company lacks a skilled technician to deal with concrete works. As a hired technician, you are requested to construct one of the sub- columns of (30*30) cm supported on a square footing of (100*100) cm and 20 cm of thickness. The work includes preparing wooden formwork and

concrete casting of the mentioned footing and sub-column within 8 hours, given that:

- ✓ The ground beam level is 1.10 m
- ✓ Concrete mix ration is 1:2:4 for footing and sub-column.
- ✓ Steel bars are already put and fixed in their place.
- ✓ All materials are available on the site.

Tools	Wire brushes, straight edge, pick axe, machete / panga, trowels, spirit level, claw hammer, squares, marker pens, chisel, hack saw, tape measure, portable circular saw, jack plane, plumb bob, clamps, screwing machines, drilling machines, claw bar, and hose pipe, broom, float, Spade/shovel and hoe.
Equipment	PPEs, wheelbarrow, concrete mixer, poker vibrator, dumpy level, metallic/plastic pans, metallic/plastic buckets, jerrycan.
Materials/ Consumables	Water, Fuel, Fine and coarse aggregates, cement, warning tape, admixtures, bricks/stones, building lines, timber, nails, lubricants, chalks, plastic sheeting and concrete spacers.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome1: Perform preliminary works. (15%)	1.1. PPE is properly used according to the work to be done.	Overall is used			0.5
		Strong boots are used			0.5
		Gloves are used			0.5
		Helmet is used			0.5
		Goggles are used			0.5
		Face mask is used			0.5
	1.2. Working /casting area is properly prepared according to the work to be done.	Working area is cleaned			0.5
		Tools are arranged			0.5
		Equipment are arranged			0.5
		Materials are arranged			0.5
	1.3. Tools are appropriately selected in accordance to the work to be done.	Mixing tools are used			0.5
		Leveling tools are used			0.5
		Measuring tools are used			0.5

		Cutting tools are used			0.5
		Fixing tools are used			0.5
		Cleaning tools are used			0.5
	1.4. Equipment is appropriately selected and used in accordance to the work to be done.	Wheelbarrow is used			0.5
		Concrete mixer is used			0.5
		Cutting machines are used			0.5
		Vibrator is used			0.5
	1.5. Materials are appropriately selected according to the work to be done.	Gravel is selected			0.5
		Sand is selected			0.5
		Cement is selected			0.5
		Water is selected			0.5
		Timbers are selected			0.5
		Nails are selected			0.5
	1.6. Concrete spacers are properly prepared according to the designed concrete cover.	Concrete spacers positioned			1
		Concrete spacers are fixed			1
Learning outcome 2: Mix concrete (20%)	2.1. Mixing ratio is properly identified according the designed concrete class/grade	Cement proportion is identified			1
		Sand proportion is identified			1
		Gravel proportion is identified			1
	2.2. Concrete materials are properly quantified based on given mixing ratio.	Cement is quantified			1
		Sand quantity is quantified			1
		Gravel quantity is quantified			1
		Water is quantified			1
	2.3. Mixing place is adequately prepared as per required standards.	Mixing place is formed			1
		Mixing place is impermeable			1
		Mixing place is clean			1
	2.4. Concrete materials are	Aggregates are piled			1

	properly collected according to their types.	No wastage			1
	2.5. Concrete materials are accurately batched in accordance with mix ratio.	Sand is batched			1
		Gravel is batched			1
		Cement is batched			1
		Water is proportioned			1
	2.6. Concrete materials are correctly mixed according to the mixing methods.	Concrete has uniform colour/ homogeneous			1
		No segregation			1
		No impurities			1
		Concrete is workable			1
Learning outcome 3: Erect formwork. (40%)	3.1. Formwork pieces are correctly prepared according to the work to be done.	Pieces are planed			2
		Pieces are measured and marked			2
		Pieces are cut			2
		Pieces are operated			2
	3.2. Formwork panels are properly prepared according to the structure element to be made.	Panel sizes are respected			2
		Panels are solid			2
		No gaps in joints			2
		Panel internal surfaces are finished			2
	3.3. Formwork panels are properly assembled according to the structure element to be made.	Footing shape is respected			2
		Footing size is respected			2
		Sub-column shape is respected			2
		Sub-column size is respected			2
		Panels are fixed			2
		Angles are right			2
	3.4. Formwork supporting structure is accurately prepared according the structure element to be made.	Formwork supporting structure is stable			2
		Formwork supporting structure is strong			2
	3.5. Formwork is tightly fixed	Footing formwork is levelled			2

	according to the designed structure.	Footing formwork is stable			2
		Sub-column formwork is levelled			2
		Sub-column formwork is stable			2
Learning outcome 4: Cast concrete (20%)	4.1. Formwork is properly assessed before concrete casting.	Formwork leakage is assessed			1
		Formwork straightness is assessed			1
		Formwork cleanliness is assessed			1
	4.2. Concrete is properly transported in accordance with transport modes.	No concrete wastage			2
		Concrete freshness is kept			2
	4.3. Concrete is properly poured and vibrated according to the standards.	No bleeding			2
		No segregation			2
		No voids			2
	4.4. Concrete surface is properly levelled according to the design.	Concrete surface is flat			2
		Concrete surface is float finished			2
	4.5. Concrete is adequately cured according to the standards.	Concrete method is applied			1
		Curing period is respected			1
		Curing time is respected			1
Learning outcome 5: Clean the workplace (5%).	5.1. Working area is properly cleaned after work.	No wastes left at the working area			1
		No scatted materials			1
	5.2. Used tools and	Used tools are cleaned			1

	equipment are properly cleaned according to their types.	Used equipment are cleaned			1
	5.3.Used tools and equipment are correctly stored in their right place.	Used tools are stored			0.5
		Used equipment are stored			0.5
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. Aramaki, T., Nakata, Y., Otsuka, S., & Ichise, K. (2022). EFFECT OF ERECTION METHODS ON FORMWORK DEFORMATION DUE TO LATERAL PRESSURE OF CONCRETE IN CONVENTIONAL WALL FORMWORK. *Journal of Structural and Construction Engineering*.
<https://doi.org/10.3130/aijs.87.990>
2. Bouharoun, S. (2016). Characterization of the interface between fresh concrete and formwork. *Journal of Civil Engineering and Management*. <https://doi.org/10.3846/13923730.2014.893912>
3. Concrete technology. (1991). *Building Research & Information*.
<https://doi.org/10.1080/09613219108727128>
4. Cusens, A. R. (1976). Concrete technology. Building and Environment.
[https://doi.org/10.1016/0360-1323\(76\)90009-3](https://doi.org/10.1016/0360-1323(76)90009-3)
5. Day, K. W. (2021). Properties of concrete. In *Concrete Mix Design, Quality Control and Specification*. <https://doi.org/10.4324/9780203967874-11>
6. J., S., & K.H., L. (2015). Sustainable Concrete Technology. *Civil Engineering Dimension*.
<https://doi.org/10.9744/ced.17.3.158-165>
7. Kekez, S., & Kubica, J. (2020). Connecting concrete technology and machine learning: Proposal for application of ANNs and CNT/concrete composites in structural health monitoring. *RSC Advances*.
<https://doi.org/10.1039/d0ra03450a>
8. Kevern, J. T. (1999). Advancements in Concrete Technology. *Concrete International-Detroit*.
9. Li, W., Lin, X., Bao, D. W., & Min Xie, Y. (2022). A review of formwork systems for modern concrete construction. In *Structures*. <https://doi.org/10.1016/j.istruc.2022.01.089>
10. Martin, D. W., & Koski, C. (2019). Active learning: Increasing construction management students' technical competencies through concrete formwork exercises. *Journal of Engineering Technology*.
11. Newman, J., & Choo, B. S. (2003). Advanced Concrete Technology: Concrete Properties. *Advanced Concrete Technology*.
12. Technology, C. (1970). Advancements in Concrete Technology. *Concrete*.